Recycling Rewards Coverage Report

Python test coverage analysis

> Both dedicated suites pass and achieve **100% statement coverage and 100% branch coverage** for `recycling_rewards.py`, exceeding the required 90% minimum.

Results summary

Suite	Test file	Passed	Statements	Branches	Coverage
Code coverage	CodeCoverageTest_xxxx_application.py	16	37/37	26/26	**100%**
Branch coverage	BranchCoverageTest_xxxx_application.py	26	37/37	26/26	**100%**

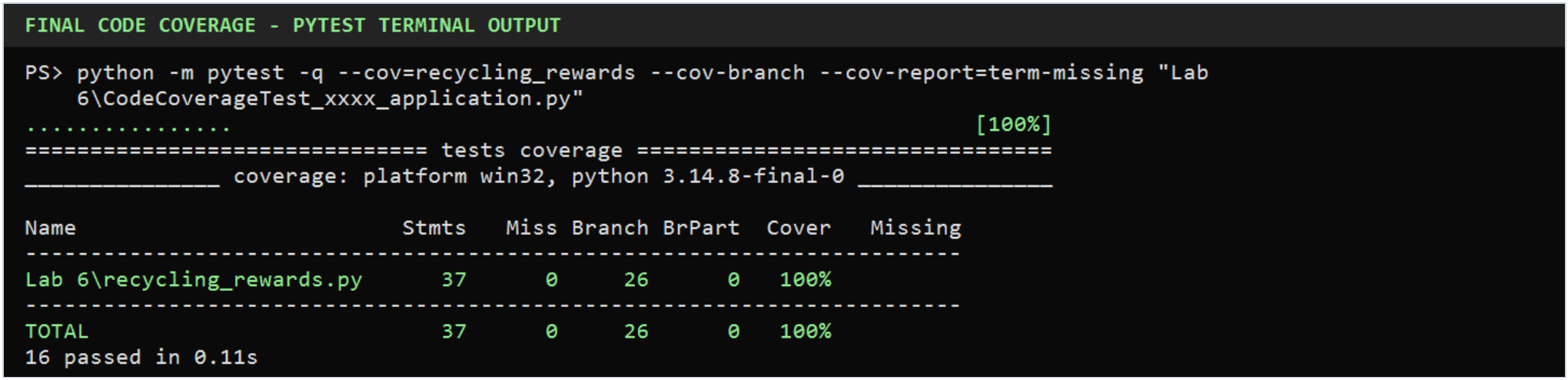
The two test files are maintained separately. Both exercise `reward_points`, `collection_accepted`, and `slot_status`. The code coverage suite checks representative behavior and invalid inputs. The branch coverage suite targets both outcomes of conditional paths, including validation, reward thresholds and rates, collection limits, and slot cutoffs.

Coverage is measured with pytest-cov and the `--cov-branch` option. The terminal output screenshots follow.

Code coverage result: 100%

16 tests passed. All 37 statements and 26 branches were covered, with no partial branches.

```
python -m pytest -q --cov=recycling_rewards --cov-branch --cov-report=term-missing "Lab 6\CodeCoverageTest_xxxx_application.py"
```



Final code coverage terminal output

Branch coverage result: 100%

26 tests passed. All 37 statements and 26 branches were covered, with no partial branches.

```
python -m pytest -q --cov=recycling_rewards --cov-branch --cov-report=term-missing "Lab 6\BranchCoverageTest_xxxx_application.py"
```

FINAL BRANCH COVERAGE - PYTEST TERMINAL OUTPUT

```
PS> python -m pytest -q --cov=recycling_rewards --cov-branch --cov-report=term-missing "Lab
    6\BranchCoverageTest_xxxx_application.py"
..........................                                               [100%]
============================== tests coverage ================================
_______________ coverage: platform win32, python 3.14.8-final-0 _______________

Name                          Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------------------
Lab 6\recycling_rewards.py       37      0     26      0   100%
-------------------------------------------------------------------------
TOTAL                            37      0     26      0   100%
26 passed in 0.12s
```

Final branch coverage terminal output